Lesson 17

Topic: Working with Maps & Geocoding, Formatting & Annotations

Prerequisite: Load `sales_with_geodata.csv` into Power BI Desktop. Ensure dataset includes:

- `Country`, `Region`, `City`
- `Latitude`, `Longitude`
- `Sales`, `Category`

Part 1: Maps & Geocoding

Task 1: Create a Basic Map Showing Sales by Country

- Visual: **Map**
- Fields:
  - Location = `Country`
  - Size = `Total Sales`
- Result: Each country is represented by a bubble, with size proportional to sales.

DAX Example:

```
Total Sales = SUM(sales_with_geodata[Sales])
```

Example:
If dataset includes USA = $1.2M, UK = $900K, Germany = $500K, bubbles will appear with different sizes to show relative sales strength.

Task 2: Use Latitude and Longitude for Precise Mapping

- Visual: **Map**
- Fields:
  - Latitude = `Latitude`
  - Longitude = `Longitude`
  - Size = `Total Sales`

Why: Avoids mismatches (e.g., “Paris” in France vs. “Paris” in Texas).

Example:

- Latitude 40.7128, Longitude -74.0060 → New York.
- Latitude 48.8566, Longitude 2.3522 → Paris.

Both cities get precise placement.

Task 3: Plot Total Sales by City

- Visual: **Map**
- Fields:
  - Location = `City`
  - Size = `Total Sales`
  - Tooltip = `Region`, `Category`, `Total Sales`

DAX Example for Category Sales:

```
Total Sales by Category =
CALCULATE(
    SUM(sales_with_geodata[Sales]),
    ALLEXCEPT(sales_with_geodata, sales_with_geodata[Category])
)
```

Example Tooltip:

- City: Los Angeles
- Region: California
- Category: Electronics
- Total Sales: $150,000

Task 4: Create a Region-wise Sales Heatmap

- Visual: **Shape Map**
- Fields:
  - Location = `Region`
  - Values = `Total Sales`
- Conditional formatting with gradient (blue = low sales, red = high sales).

Example:

- East Region = $2.5M (red)
- West Region = $1.2M (orange)
- South Region = $700K (yellow)
- North Region = $300K (blue)

Quickly identifies top/bottom-performing regions.

Task 5: Add Drill-Down from Country → Region → City

- Build hierarchy: `Country → Region → City`.
- Apply hierarchy in **Map visual**.
- Enable **Drill Down** mode.

Example Usage:

- Start with Country = USA → Drill to Region = California → Drill to City = Los Angeles.

Part 2: Formatting & Annotations

Task 6: Customize Map Visual

- Title: *"Global Sales Performance"*
- Theme: Dark background with bright-colored bubbles.
- Enable data labels for clarity.

Professional Tip: Always keep map readable; avoid clutter with too many labels.

Task 7: Use Conditional Formatting for Bubble Colors

- Go to **Data Colors → Conditional Formatting**.
- Base on `YoY Growth %`.

DAX Example:

```
YoY Sales =
CALCULATE(
    SUM(sales_with_geodata[Sales]),
    DATEADD(sales_with_geodata[Date], -1, YEAR)
)

YoY Growth % =
DIVIDE([Total Sales] - [YoY Sales], [YoY Sales], 0)
```

Color Scheme:

- Green: Growth > 10%
- Yellow: 0–10%
- Red: Negative growth

Task 8: Add Annotations to Highlight Key Cities

- Insert **Shapes/Text Boxes** → highlight top markets.
- Example annotation: *"New York – Top Sales: $1.5M (+12% YoY)"*.

Professional Tip: Use annotations sparingly for storytelling impact.

Task 9: Create a Tooltip Page with Detailed Info

- New Page → mark as **Tooltip**.

- Add visuals:
  - **Card:** `Total Sales`
  - **KPI:** `YoY Growth %`
  - **Bar Chart:** Sales by Category

Assign tooltip page to map visual → hover over a city shows detailed info.

Example: Hovering over *Paris* shows:

- Total Sales = $900K
- YoY Growth = +8%
- Top Category = Luxury Goods

Task 10: Add a Slicer to Filter the Map by Category

- Insert **Slicer** → Field = `Category`.
- Interaction: When selecting *Electronics*, map updates to show only Electronics sales across geographies.

Professional Tip: Combine slicers with bookmarks to create dynamic map reports.